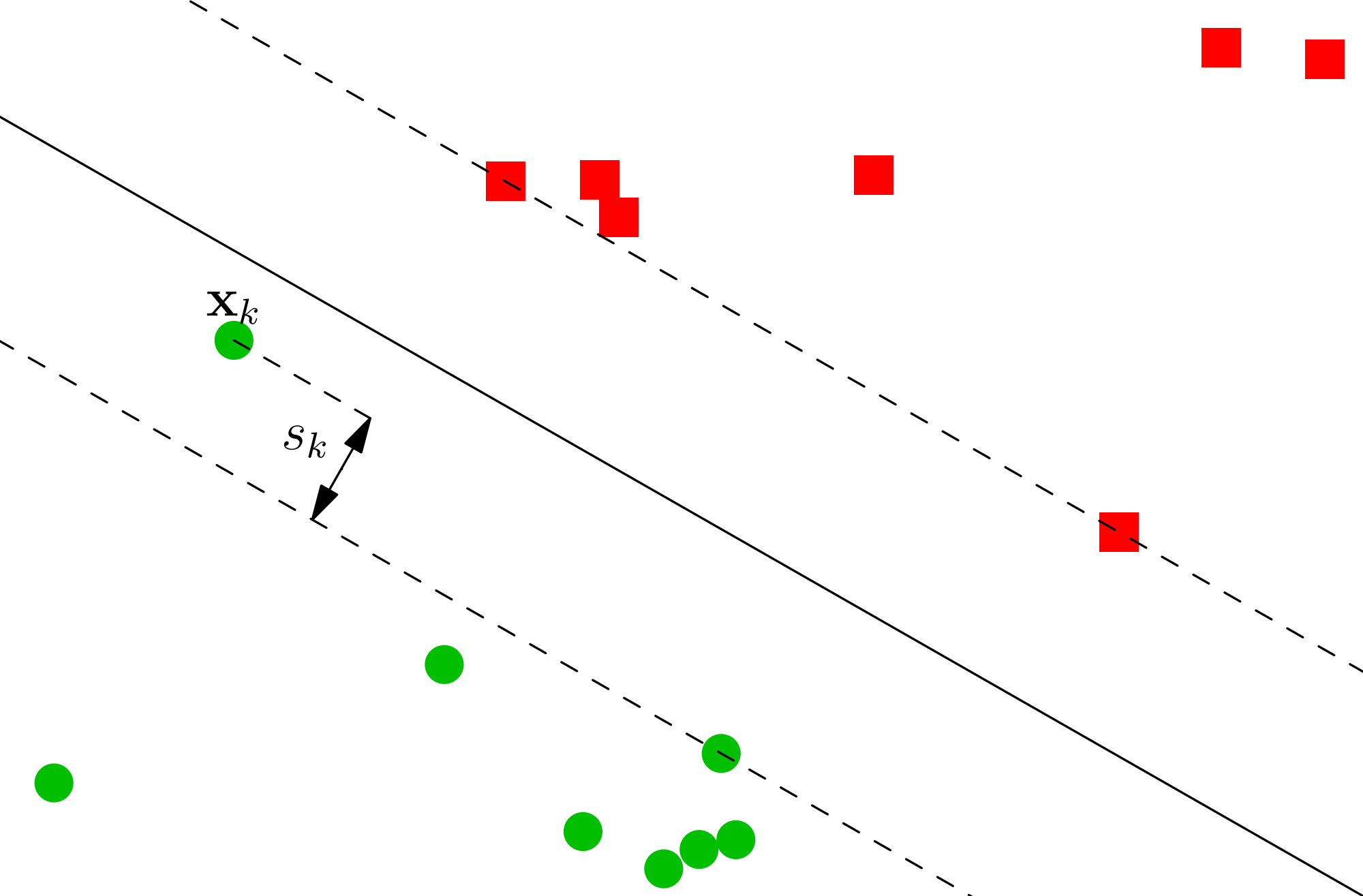
$\mathbf{x}_k$
$s_k$